✅ Simplification ও Karnaugh Map

1. Simplify F(w, x, y, z) = Σ(1, 3, 7, 11, 15), with don't-care d(w, x, y, z) = Σ(0, 2, 5), and show that an expression with the minimum number of literals is not necessarily unique. ★
2. Simplify the Boolean function by using the **tabulation method**:
   F = Σ(0, 1, 2, 8, 10, 11, 14, 15)
3. Simplify the Boolean function f(A, B, C, D) = Σ(0, 1, 2, 3, 5, 7, 8, 10, 12, 13, 15) using the tabulation method. ★
4. Simplify the Boolean function F(w, x, y, z) = Σ(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15)
5. Simplify F = A'B'C'D + B'CD' + A'BCD' + AB'C' and implement the result using logic gates. ★
6. Simplify the Boolean function F = A'B'C'D + B'CD' + A'BCD' + AB'C' and implement using logic gates. ★
7. Simplify the Boolean function F(w, x, y, z) = Σ(1, 3, 7, 11, 15) with don't care d(w, x, y, z) = Σ(0, 2, 5) — show "An expression with the minimum number of literals is not necessarily unique". ★
8. F(A, B, C, D) = Σ(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15) — Show prime implicants and 7 essential prime implicants using Karnaugh map.

✅ De Morgan's Theorem ও Complement

9. Find the complement of the function Fi = xyz' + x'y'z' and F = (yz' + yz) applying **De Morgan's Theorem**. ★
10. Find the complement of the function Fi = xyzi + x'y'z' and F = (yz + yz') applying De Morgan's Theorem.
11. What do you mean by complements? Make a comparison between 1’s and 2’s complements.

✅ Boolean Proofs ও Maxterm

12. Prove that fi = $m_0 + m_1 + m_6 + m_7 = M_0.M_3.M_4.M_5$ ★
13. Prove that fi = $m_0 + m_1 + m_6 + m_7 = M_2.M_3.M_4.M_5$
14. Express the following function in **product of maxterm** form:
    f(w, x, y, z) = y'z + wxz' + w'x'z

✅ Theoretical Concepts ও Code Conversion

15. "An expression with the minimum number of literals is not necessarily unique" — illustrate with example. ★
16. What are the **advantages and disadvantages** of digital techniques over analog techniques?

~~17. What~~ is **duality principle**? Prove $x + x \cdot y = x$ by duality.

18. Represent the decimal number 8620:

- (i) in **biquinary**
- (ii) in **excess-3 code**
- (iii) in **2-4-2-1 code**

~~19. Con~~vert octal **623.77** to:

- decimal
- binary
- hexadecimal

20. Convert $(511)_4$ to base-5.

✅ Logic Gate ও Circuit প্রশ্নসমূহ

21. Draw **switching circuits** for 2-input NAND, NOR, and AND operations.
22. Draw and describe **4-bit Bidirectional shift register with parallel load**

✅ Mixed/Complex Boolean Expressions

23. Simplify the Boolean function F using the don't-care conditions:
    $F = w(x'y + x'y' + xyz) + x'z'(y + w)$
    $d = w'x(y'z + yz') + wyz$